

states

`on-table(x)` · `on(x, y)`
`clear(x)` · `holding(x)`
`hand-empty`

actions

`pick-up(x)` · `unstack(x, y)`
`put-down(x)` · `stack(x, y)`

preconditions + effects

`pick-up(x)`: `on-table(x)` $\wedge$ `clear(x)` $\wedge$ `hand-empty`
 $\rightarrow$ `holding(x)` $\wedge$ $\neg$ `on-table(x)` $\wedge$ $\neg$ `hand-empty`

initial state + goal

`on(blue, orange)` · `clear(red, blue, yellow)`
goal: `on(orange, blue)`

plan

`unstack(blue, orange)`
`put-down(blue)`
`pick-up(orange)`
`stack(orange, blue)`